

Automatidata | Executive Summary

AUTOMATIDATA | EXECUTIVE SUMMARY

Project focus

An end-to-end data science case study using an educational NYC taxi dataset. The analysis combines statistical testing, exploratory analysis, regression modeling, classification, model diagnostics, and business interpretation.

Business questions

- Do observed fares differ between credit-card and cash payment groups?
- Can trip-level information estimate taxi fares?
- Can trip characteristics identify generous-tipping trips?

Data

22,699 rows and 18 columns. No missing values and no duplicate rows were found. Initial quality checks identified negative values in several monetary fields and one negative duration; the notebook documents the cleaning rules and verifies the cleaned state.

Payment-method analysis

Welch's two-sample test estimated a credit-card-minus-cash mean fare difference of approximately \$1.22, with a 95% confidence interval of approximately \$0.87 to \$1.56 and $p \approx 6.8 \times 10^{-12}$. A positive-fare sensitivity analysis produced a similar estimate. This is an observational association, not evidence of causation.

Fare estimation

Four models were compared. Gradient Boosting was selected using validation MAE. On the untouched test set:

MAE = \$3.55

RMSE = \$16.41

$R^2 = 0.182$

The largest errors occur in the highest-fare segment, with MAE around \$9.73 for fares above \$16.

Generous tipping

Random Forest was selected using cross-validated F1. On the held-out test set:

Accuracy = 0.603

Precision = 0.456

Recall = 0.731

F1 = 0.562

ROC AUC = 0.645

The confusion matrix contains 778 TP, 286 FN, 927 FP, and 1,062 TN.

Portfolio takeaway

The project demonstrates an end-to-end workflow with explicit data-quality checks, validation-based model selection, untouched final testing, statistical sensitivity analysis, model diagnostics, and documented limitations. The results are presented as a portfolio case study rather than a production-ready taxi pricing or customer decision system.